CMPE 125 Lab 1
Light Controller Circuit

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1 Objective

The objective of this experiment is to implement and simulate a simple combinational light controller circuit using Verilog HDL. The circuit has two inputs, x_1 and x_2, and one output, f.

The output is high when the two inputs have different logic values and low when the two inputs have the same logic value.

2 Boolean Expression

From the given light controller circuit, the Boolean expression is

$$f = x_1\overline{x_2} + \overline{x_1}x_2 \tag{1}$$

This expression is equivalent to an XOR operation:

$$\boxed{f = x_1 \oplus x_2} \tag{2}$$

Therefore, the output is equal to logic 1 only when x_1 and x_2 have different values.

3 Truth Table

The expected truth table is shown below.

Table 1: Truth table of the light controller.

x_1	x_2	f
0	0	0
0	1	1
1	0	1
1	1	0

4 Verilog Implementation

The light controller was implemented in Verilog using the following Boolean expression:

```
module light(
    input x1,
    input x2,
    output f
);

    assign f = (x1 & ~x2) | (~x1 & x2);

endmodule
```

5 Testbench

A testbench was created to test all four possible combinations of x_1 and x_2.

The input sequence used in the simulation was

$$00 \rightarrow 01 \rightarrow 10 \rightarrow 11.$$

Each input combination was applied for 10 ns.

6 Simulation Results

The behavioral simulation result from Vivado is shown in Figure 1.

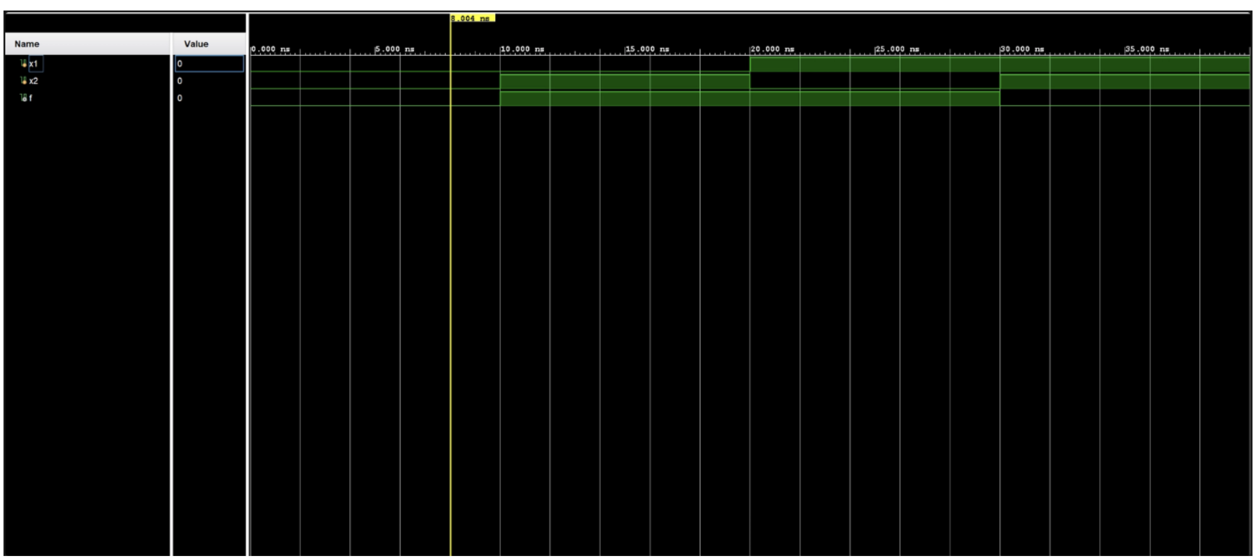

Figure 1: Behavioral simulation waveform of the light controller.

The simulation results are summarized below.

Table 2: Simulation results.

Time (ns)	x_1	x_2	Expected f	Simulated f
0–10	0	0	0	0
10–20	0	1	1	1
20–30	1	0	1	1
30–40	1	1	0	0

The simulated output sequence was

$$\boxed{0,\ 1,\ 1,\ 0}$$

which matches the expected truth table.

7 Discussion

The simulation results match the theoretical results for all four input combinations. When x_1 and x_2 have the same value, the output is 0. When x_1 and x_2 have different values, the output is 1.

There were no discrepancies between the expected values and the simulation results.

8 Conclusion

The light controller circuit was successfully implemented and simulated using Verilog HDL in Vivado. The circuit implements the Boolean function

$$f = x_1\overline{x_2} + \overline{x_1}x_2$$

which is equivalent to an XOR operation. The simulation produced the output sequence $0, 1, 1, 0$, which matched the expected truth table for all possible input combinations.